

GABORONE, BotswanaWMO: **682400**Lat: **24.555S**Lon: **25.918E**Elev: **1006**StdP: **89.81**Time Zone: **2.00 (E02)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.9	3.2	-14.1	1.2	18.0	-12.1	1.5	17.4	9.0	18.3	8.0	18.8	1.8	270	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	36.9	16.9	35.5	17.0	34.2	17.1	21.7	26.3	21.2	26.0	20.7	25.9	4.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	17.6	22.4	20.1	16.8	22.2	19.8	16.4	22.2	68.4	26.4	66.8	25.4	65.0	26.2	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.9	DB	-1.6	38.9	1.9	1.4	-3.0	39.9	-4.0	40.7	-5.1	41.5	-6.4	42.6	(4)
(5)			WB	-4.5	22.8	1.4	0.3	-5.5	23.0	-6.3	23.2	-7.1	23.4	-8.1	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	25.7	25.5	23.7	20.3	16.5	13.8	13.4	16.7	21.0	24.2	24.9	25.4
	DBStd	5.22	2.36	2.34	2.31	2.46	2.49	2.17	2.31	2.93	3.03	3.05	2.97	2.60
	HDD10.0	6	0	0	0	0	1	2	3	1	0	0	0	0
	HDD18.3	447	0	0	1	9	66	137	154	67	11	1	1	0
	CDD10.0	3987	486	435	426	311	203	115	107	209	329	441	448	478
	CDD18.3	1387	227	202	168	69	9	1	0	17	90	184	199	220
	CDH23.3	15109	2227	1998	1513	656	301	76	81	410	1239	2192	2181	2236
	CDH26.7	6739	1042	939	628	191	54	6	6	103	539	1094	1081	1056
Wind	WSAvg	3.1	3.3	3.1	2.8	2.6	2.4	2.4	2.6	3.2	3.6	4.0	3.9	3.5
Precipitation	PrecAvg	431	80	65	62	24	12	8	1	1	5	32	64	72
	PrecMax	838	200	242	168	138	92	96	10	6	63	110	276	207
	PrecMin	213	6	0	2	0	0	0	0	0	0	1	6	22
	PrecStd	154	62	65	49	30	19	19	2	2	13	29	53	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	37.1	36.2	32.8	30.8	27.2	27.2	31.2	35.2	37.8	38.2	37.2
		MCWB	17.4	17.7	16.8	16.0	13.7	12.0	11.5	13.8	15.1	16.0	16.7	17.7
	2%	DB	35.2	35.2	33.9	30.8	28.0	25.2	25.4	29.3	33.3	35.9	36.1	35.5
		MCWB	18.0	18.1	17.8	16.6	13.4	11.7	10.8	13.0	14.4	15.7	16.4	17.8
	5%	DB	33.9	33.8	32.0	29.0	26.5	23.9	23.9	27.8	32.0	34.1	34.2	33.9
		MCWB	18.2	18.3	17.6	16.2	13.1	11.2	10.5	12.4	14.4	15.5	16.9	18.1
	10%	DB	32.0	32.0	30.1	27.2	25.0	22.1	22.0	25.8	30.1	32.2	32.8	32.1
		MCWB	18.5	18.6	17.9	16.2	13.0	11.2	10.7	12.0	13.4	15.5	16.5	18.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	22.0	21.7	20.1	16.7	14.4	13.6	15.6	17.4	19.3	20.8	21.9
		MCDB	26.8	25.9	26.4	24.2	23.7	22.1	22.2	27.6	28.5	28.2	26.5	27.3
	2%	WB	21.7	21.4	21.2	19.2	15.9	13.4	12.7	14.6	16.5	18.6	20.1	21.2
		MCDB	26.2	25.9	25.7	24.7	23.4	21.5	21.4	26.3	27.7	27.9	26.4	26.8
	5%	WB	21.2	20.9	20.6	18.6	15.2	12.6	11.9	13.9	15.9	18.0	19.5	20.6
		MCDB	25.8	25.9	25.0	24.5	23.2	21.2	20.7	25.3	27.4	27.1	26.6	26.5
	10%	WB	20.7	20.5	20.1	18.0	14.2	11.8	11.1	13.1	15.1	17.3	18.9	20.1
		MCDB	25.9	25.9	24.3	23.7	22.5	21.0	20.5	23.7	27.0	27.0	26.8	26.3
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.6	12.0	13.1	16.7	17.3	17.8	17.6	16.6	14.6	13.0	11.8
		MCDBR	14.5	14.7	15.3	15.8	19.0	20.2	20.5	19.9	19.0	17.3	15.7	14.5
		MCWBR	3.8	4.3	4.5	5.1	7.9	9.2	9.8	8.2	7.5	6.1	4.9	3.9
	5% WB	MCDBR	10.3	10.5	10.4	11.4	15.2	16.5	16.9	16.9	15.5	14.1	12.4	10.9
		MCWBR	3.4	3.5	3.8	4.2	6.6	8.0	8.7	7.1	6.9	5.5	4.4	3.8
Clear-Sky Solar Irradiance	taub	0.382	0.389	0.372	0.344	0.302	0.287	0.286	0.348	0.397	0.421	0.372	0.375	
	taud	2.389	2.364	2.396	2.433	2.513	2.525	2.512	2.333	2.202	2.186	2.371	2.404	
	Ebn at Noon	962	940	927	910	913	908	922	892	888	899	964	970	
	Edh at Noon	129	130	120	106	89	83	87	115	143	153	131	128	
All-Sky Solar Radiation	RadAvg	6.95	6.43	5.82	4.97	4.57	4.09	4.44	5.23	6.19	6.75	7.00	7.02	
	RadStd	0.68	0.61	0.54	0.45	0.19	0.21	0.13	0.23	0.27	0.35	0.66	0.48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page